

AURÉLIEN FEIGNON



CONTACT

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Tours, France

EDUCATION

APPLICATION DESIGNER & DEVELOPER

2022 · ENI School

WEB & MOBILE WEB DEVELOPER

2021 · ENI School

BACCALAURÉAT (FRENCH HIGH-SCHOOL DIPLOMA)

2011

SKILLS

AI & AGENTS

LLM integration (Anthropic, OpenAI, Ollama)

· Tool-calling / function calling · Structured outputs & schema validation · Autonomous agent design · Claude Code (skills, agents, hooks, MCP) · Prompt & context engineering

BACKEND & ARCHITECTURE

TypeScript · Node.js (Express, WebSocket) · Python · Symfony / API Platform · Prisma ORM · Pydantic · RabbitMQ · Redis · Mercure · Microservices · Event-driven architecture

DATA & ML

PostgreSQL · TimescaleDB · MySQL · TensorFlow / Keras · XGBoost · NVIDIA Triton · CUDA / GPU · Time series

FRONTEND & INFRASTRUCTURE

React · Next.js · Zustand · React Flow · Docker · Traefik · GitHub Actions / CI · Cloudflare Tunnel · MiniIO (S3) · Grafana

LANGUAGES

- French — native
- English — B2

INTERESTS

- Sport: skiing, swimming, body pump
- Author of texts and screenplays
- Animation, screen printing, embroidery

PROFILE

Senior Full Stack Developer specialised in real-time architectures and distributed systems, with hands-on experience deploying ML pipelines (Python/TensorFlow) and agentic LLM features (tool-calling, multi-provider). I work from architectural design to operations, with a strong focus on reliability, traceability and performance.

KEY EXPERTISE

- Microservices architectures
- Event-driven systems
- Real-time applications
- LLM agent engineering (tool-calling, structured outputs)
- ML pipelines in production
- Documented architecture (RFC, ADR)
- Reliable distributed systems (idempotency, recovery)
- Strong technical autonomy

PROFESSIONAL EXPERIENCE

AUGURE — REAL-TIME PREDICTIVE PLATFORM

2025 – Present

TypeScript/Python monorepo — 48 containerised services, event-driven architecture with an integrated AI assistant.

- Product-integrated agentic conversational assistant:** tool-calling orchestrator exposing **27 business tools** to the model (predictive-model creation, network-architecture editing, training launches, strategy analysis and optimisation), **streaming** responses
- LLM multi-provider abstraction:** unified client interface with **Anthropic**, **OpenAI** and **Ollama** (self-hosted) adapters; API keys **encrypted at rest**, **SSRF** protection on configurable endpoints
- AI tool security:** every tool enforces the application's visibility rules and state guards — the model only accesses resources the user is authorised for, never beyond
- Event-driven microservices** architecture: ~30 specialised workers orchestrated by **RabbitMQ** (job queues, DLQ, supervision, idempotent retries)
- End-to-end ML pipeline:** ingestion → features → training with **TensorFlow/ Keras** (multi-head regression + classification models) and **XGBoost** → GPU inference via **NVIDIA Triton Inference Server**
- Agentic research pipeline** (Python): agents with **validated structured outputs** (Pydantic), a provider abstraction enabling a testable deterministic mode, and a graduated autonomy scale with mandatory human validation before any deployment
- Architecture RFC design** (35 chapters, **169 tracked decisions**) turning the platform into an auditable research lab, with automated documentation-compliance checking

- **Cross-language typed contracts:** **Prisma** → **JSON Schema** → **Pydantic** generation, with structural CI guards enforcing TypeScript/Python parity
- **Large-scale time series:** **PostgreSQL/TimescaleDB** (151 models, hypertables), real-time multi-provider ingestion (market WebSockets, option chains, news feeds)
- **Security & multi-tenant:** least-privilege PostgreSQL role separation, **JWT** + **Argon2**, inter-service HMAC, organisations, roles and resource visibility
- **Reliability & operations:** backtesting and rolling-KPI engine, proven backup/restore chain, **Grafana** observability, remote access via **Cloudflare Tunnel/Access**
- **CI/CD:** self-hosted GitHub runner, unified test gate (**vitest** + **pytest**), 214 versioned migrations
- **Docker** / **Traefik** reverse proxy, **typed WebSocket router** for real-time delivery, real-time client-side visualisation

○ **ASKOR — LCA & ECO-DESIGN SOFTWARE**

2021 – Present

- Design of a **component library** and complex dynamic workflows
- Interactive modelling with **React Flow** and advanced state management (Zustand)
- API structuring and evolution via **Symfony API Platform**
- Development of an asynchronous computation engine (**Symfony Messenger** + **Python microservice**) with message-queue handling
- Real-time streaming via **Mercure**
- Design of a dynamic eco-design module (**weighted decision tree**)
- Optimisation and refactoring of a multi-sector LCA (refactoring, service **lazy-loading**)
- Versioning of emission factors and **result persistence**
- Data visualisation with **Highcharts**